



Humza Asrar Ahmed

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ABOUT ME

Software Engineer & aspiring Earth System Data Scientist striving to blend technical acumen with geospatial intelligence to solve socio-environmental challenges. With a track record of crafting secure systems and AI-driven spatial analytics, I bring a nuanced understanding of how technology intersects with societal needs—from safeguarding data infrastructures to interest in optimizing GIS frameworks. Recognized for bridging analytical rigor with creative problem-solving, I thrive on transforming complex problems into scalable, ethical solutions. Passionate about research that pioneers AI's role in environmental and geospatial sustainability, I aim to redefine STEM's real-world impact—one certified algorithm, one spatial dataset at a time.

WORK EXPERIENCE

LIDAR DATA ANALYST (INTERN) – JULIUS KÜHN-INSTITUT – 07/2026 – 09/2026 – QUEDLINBURG, GERMANY

- Built a Python pipeline (laspy, NumPy, SciPy) that localizes, measures and segments 444 individually surveyed conifer seedlings from drone LiDAR, from raw point clouds to crown polygons
- Compared seedling height across 7 flight-altitude combinations (30 m / 60 m, single and merged) using Wilcoxon tests; showed 60 m data alone underestimates height
- Processed TLS and UAV-LiDAR point clouds of up to ~27 million points (ground classification, canopy height models, foliage height diversity) and related them to LAI measurements - Implemented and cross-validated watershed and DeepForest tree detection
- Diagnosed and fixed memory failures and a normalization bug; documented methods and limitations for reproducibility
- Flew UAV missions (LiDAR + RGB) and verified and corrected tree positions in QGIS - Supported VOC sampling on Norway spruce under artificial drought (20 trees) and a Diplodia infection experiment on pine saplings under normal vs. water stress

BUSINESS ANALYST – DANSAN LLC – 17/02/2025 – 01/10/2025 – SAN RAMON, UNITED STATES

- Utilized advanced analytics to monitor and optimize Amazon product pricing, sales trends, and inventory levels.
- Collaborated cross-functionally to blend market intelligence and customer insights, refining pricing strategies and promotional campaigns for peak product performance.

JUNIOR AI ENGINEER (CONTRACT) – QUANTUM PARADIGM – 27/01/2025 – 07/03/2025 – KARACHI, PAKISTAN

- Developed an automated ID card processing pipeline integrating YOLOv8 for face detection, portrait extraction, and OCR (Tesseract & EasyOCR) for comprehensive text capture.
- Engineered algorithms using regex, layout parsing, and deep learning to accurately match attribute labels with values, including multilingual support (e.g., Urdu transcription).
- Optimized system performance through rigorous debugging and documentation, ensuring precise segmentation and maintainable code architecture.

FULL-STACK DEVELOPMENT INTERN – INFINITE ALGOS – 02/07/2024 – 11/10/2024 – ISLAMABAD, PAKISTAN

- Spearheaded full-stack development and project planning, incorporating requirement gathering, user-centric design, secure authentication/authorization (including sign-up/sign-in), robust database design, and RESTful API integration using Express.js and React.js.
- Delivered a cohesive and secure solution by implementing industry best practices for data handling and credential hashing, ensuring seamless interaction between frontend and backend systems for an enhanced user experience.

LMS IMPLEMENTATION ENGINEER – FORMATIVE ACADEMY OF HIGHER EDUCATION AND MANAGEMENT – 12/07/2024 – 18/07/2024 – KARACHI, PAKISTAN

- Deployed and configured a Learning Management System on Hostinger, ensuring seamless Moodle functionality.

PROCESS ASSOCIATE – CLYDE CHEMICALS – 26/06/2024 – 18/07/2024 – KARACHI, PAKISTAN

• Acquired hands-on analytical and process optimization skills through intensive observation of manufacturing operations, enhancing data-driven decision-making for STEM innovation.

QUALITY ASSURANCE ASSOCIATE INTERN – NEODOCTO INC – 01/04/2023 – 28/06/2023 – NEW YORK, UNITED STATES

- Automated e-commerce product listings and geospatial data extraction from Google Maps using Python (Selenium & BeautifulSoup) for enhanced accuracy and GIS insights.
- Optimized digital operations by integrating advanced project management tools with contemporary digital marketing strategies.

WEBSITE DEVELOPER & CONTENT WRITER – PEARL INTERNATIONAL – 10/07/2022 – 10/08/2022 – ISLAMABAD, PAKISTAN

- Designed and developed a responsive product listing website using HTML, CSS, and JavaScript, ensuring detailed and coherent content through rigorous research and expert collaboration.

ASSOCIATE EDITOR & RESEARCH INTERN – INSTITUTE OF POLICY STUDIES – 01/07/2021 – 02/08/2021 – ISLAMABAD, PAKISTAN

- Authored and published a study in Policy Perspectives on youth sports culture, societal benefits, and district-level policy measures.
- Optimized multilingual (English/Urdu) audio-to-text transcription using SaaS automation and manual refinements, bolstering academic research.

● **EDUCATION AND TRAINING**

06/10/2025 – CURRENT Leipzig, Germany
MASTERS IN EARTH SYSTEM DATA SCIENCE & REMOTE SENSING Leipzig University

- Applied AI/ML to geodata and built end-to-end spatio-temporal workflows following FAIR data principles
- Analyzed satellite climate products (soil moisture, albedo, land cover) and hyperspectral, radar and LiDAR data, with field sampling for ground truthing
- Numerous Projects as part of coursework

Website uni-leipzig.de |

Field of study Earth sciences , Environmental sciences , Natural environments and wildlife , Research Data Management & Social Responsibility, Applications of Remote Sensing Products, Global Remote Sensing Data Products, Advanced Data Analytics, Ground Truthing, Spatio-Temporal Data, Scientific Writing, Applied Topics in Earth System Science, Imaging & Non-imaging Reflectance Spectroscopy , Microwave & LiDAR Remote Sensing, Applied Data Analysis of Earth Surface Processes, Geology of Cenozoic Era

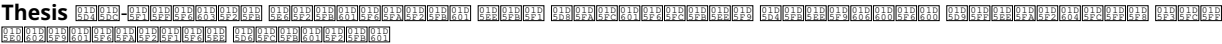
Level in EQF EQF level 7 | **National classification** 7

11/10/2021 – 11/06/2025 Haripur, Pakistan
BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology

- Solid grounding in data structures, AI/ML, advanced network security, with hands-on security firewall and Linux scripting, applying to real-world GIS applications.
- Gained industry exposure through field visits and collaborative projects, seamlessly bridging academic knowledge with practical solutions in cybersecurity and geospatial data.
- Actively participated in Model OIC debating society as a representative, environmental initiatives, and GIS-focused events, honing communication skills and interdisciplinary expertise.

Field of study Discrete Structures (COMP-121), Applied Physics (PHY-101), Calculus and Analytical Geometry (MTH-101), Object Oriented Programming (COMP-112), Probability and Statistics (MTH-210), Data Structures and Algorithms (COMP-201), Operation Research (COMP-223), Linear Algebra (MTH-204), Operating Systems (COMP-262), Database Systems (COMP-231), Software Requirement Engineering (COMP-271), Mobile Application Development (COMP-313), Simulation and Modeling (COMP-323), Computer Networks (COMP-352), Human Computer Interaction (COMP-272), Software Design and Architecture (COMP-273), Software Quality Engineering (COMP-380), Software Construction and Development (COMP-370), Information Security (COMP-291), Software Project Management (COMP-470), Machine Learning (COMP-240), Computer Vision (COMP-440), Intellectual Property Law (MGT-356), Software Testing Tools and Techniques (COMP-385)

Level in EQF EQF level 6 | **National classification** PNQF Level 6 | **Type of credits** CHS | **Number of credits** 130 |

Thesis 

09/2019 – 07/2021
IB DIPLOMA International Baccalaureate

- Cultivated advanced analytical and research skills through a rigorous IB curriculum, with a strong emphasis on STEM subjects and interdisciplinary projects.

• Enhanced leadership and teamwork by actively engaging in debate, adventure, and environmental initiatives, balancing academic excellence with community involvement.

Field of study German AB SL, Psychology SL, Chemistry HL, Physics HL, Mathematics Analysis and Approaches HL, Theory Of Knowledge , Extended Essay, English A: Lang and Literature SL

Final grade 33 | **Level in EQF** EQF level 4 | **Type of credits** IB Diploma Points Framework | **Number of credits** 45 |

Thesis 


10/2017 – 06/2019 Karachi, Pakistan
SECONDARY SCHOOL CERTIFICATE (MATRICULATION) Army Public School & College System

• Demonstrated leadership as a Model UN delegate and basketball captain, while deepening scientific inquiry beyond the standard Matric curriculum through rigorous self-study and mentorship from a distinguished theoretical physics researcher.

Field of study Physics, Chemistry, Mathematics, Biology, English, Urdu, Islamiyat, Pakistan Studies | **Final grade** A | **Level in EQF** EQF level 3

● **SKILLS**

Software Development | Retrieval-Augmented Generation | Market Analysis & Planning | Competitive Intelligence | Strategic Problem Solving | Creativity & Innovation | Strategic & Corporate Communication | Large Language Models Operations (LLMOps) | Artificial Neural Networks | AI | Docker | Git | Python | JavaScript | Research & Development (R&D) | Linear Programming | Analytical Reasoning & Method Development | Kali Linux | SQL | Docker & Kubernetes | Data-Structures & Algorithms | Machine Learning & Vision | Encryption & Cryptography | Terminal & Shell Operations (CLI) | DB & Web Administration | Continuous Improvement | Computer Networking & Security | Process Simplification | Geospatial software - ArcGIS, Land Change Moduler, Arc SWAT, SWAT CUP | Other skills: Data-Science, GIS and Remote Sensing, Math and Statistic | FOSS tools for geospatial analysis

● **PROJECTS**

10/2025 – 03/2026
Algae bloom South Australia 2025

Tracking water surface temperatures, chlorophyll a and spectral indices before & after March 2025

10/2024 – CURRENT
AI-driven Sentiment and Emotional Analysis Framework for Multimedia Content

• Developed a robust NLP pipeline using transformer models (RoBERTa, BERTopic), spaCy, and custom text chunking with dynamic Matplotlib/Streamlit visualizations for real-time transcript analysis and actionable insights.

01/2025 – 26/02/2025
Gatsby React Portfolio Website

- Engineered a high-performance static website using Gatsby, leveraging GraphQL for dynamic content, styled-components for modular styling, and continuous Netlify deployment—featuring optimized images, SEO best practices, and smooth animations for an engaging user experience

12/2023 – 01/2024
Proxy Firewall for Secure Traffic via Public VPS IP Masking

• Configured Pfense proxy server with Squid for reverse proxy caching, implemented pfBlockerNG for mass DNS blocking, and deployed Ntopng for comprehensive network monitoring and logging.

05/2023 – 06/2023
Task Automation using AutoHotKey.ahk Scripting

• Streamlined repetitive GUI tasks using AutoHotkey by mapping shortcuts for frequent texts and special characters, and compiling scripts into executables for universal deployment.

Link <https://github.com/Drugoy/Autohotkey-scripts-.ahk>

12/2022 – 03/2023
Airplane Seat, Passenger and Aircraft Deck Allocation System

• Engineered an airplane seating and deck allocation system leveraging OOP and advanced data structures (queues, linked lists, constructors, pointers, etc.) for optimized resource management.

Link https://github.com/HumzAhme/Airplane-Seat_Deck-Reservation-System

06/2022 – 08/2022
Web & Content Developer (HTML, CSS, JS ; WordPress)

- Developed a dynamic website using HTML, CSS, and JS featuring custom product content, an integrated client email subscription database and hosting.
- Link <https://pearl-intl.com/>
- 02/2022 – 03/2022
- Algorithmic Text Encryption Engine (C++)**
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- Developed a CLI-based tool that leverages a data-driven Caesar cipher—using arithmetic on ASCII codes with a user-specified key—to encrypt and decrypt text strings, highlighting algorithmic design and mathematical precision.
- Link <https://github.com/HumzAhme/1st-encryptionDecryption>

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C2
GERMAN	A2	A2	A2	A2	A2
ARABIC	A2	A2	A1	A1	A2
PANJABI; PUNJABI	B2	B2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **VOLUNTEERING**

08/2026 – 08/2026 Leipzig

Transition & Athlete Services Volunteer | IRONMAN 70.3 Leipzig

- Supported athlete check-in and race-weekend logistics at the inaugural IRONMAN 70.3 Leipzig (around 2,500 athletes).
- Handled morning clothes bags, checked bike racking and bag placement in transition, briefed international athletes on the race route in English at the Info Desk, and activated RFID timing chips at the check-in exit.

06/2026 – 06/2026 Leipzig

Guest Services & Access Volunteer | MACHN'26 Festival

- Front-of-house volunteer at a two-day business, technology and culture festival with 70+ speakers across five stages.
- Managed check-in and ticket scanning for multiple ticket categories, acted as a communication link between organisers, volunteers and attendees, and covered stages and workshop rooms to keep sessions running smoothly.

12/01/2025 – 26/01/2025 Islamabad

IRU Marathon Course Guide | Islamabad Run with Us

- Provided real-time course guidance for over 3,000 marathon participants using effective verbal, nonverbal, and multilingual communication to ensure safe navigation.
- Collaborated with team leads on logistics and RFID tracking, enhancing leadership and teamwork skills through coordinated on-course support.
- Boosted event readiness by assisting with pre-race setup and providing timely refreshment support during high-traffic segments.

06/12/2024 – 08/12/2024 Islamabad

Commentator, Performance Tracker, Coordinator, Volunteer Staff | Islamabad Cycling Association

- Provided bilingual (Urdu/English) live commentary and motivational support, enhancing the spectator and cyclist experience.
- Managed event logistics and performance tracking—including video documentation and real-time position monitoring—to ensure seamless event execution.

- Facilitated interactive workshops for 9th-grade students, focusing on digital literacy and financial independence through engaging activities, brainstorming sessions, and hands-on business simulations, fostering critical thinking and entrepreneurial skills.
- Developed and implemented practical learning experiences that encouraged students to calculate costs, understand profit margins, and learn from real-world business scenarios, promoting integrity and a growth mindset.
- Conducted empathy interviews of struggling working class of Pakistan.

● **HOBBIES AND INTERESTS**

Sports Connoisseur

Rock Climbing | Running | Skiing | Ice Skating | Long Tennis | Swimming | Horse Riding | Taekwondo | Football | Gymnastics

Personal Interests

Geospatial Storytelling & Mapping | Animal Welfare | Knowledge Sharing | Philosophical Debates | Societal Discourse | Cross-cultural Research using GIS tools | Wildlife Preservation | Children Education

● **CERTIFICATIONS & LICENSES**

08/2024 – 08/12/2024

Generative AI Application Developer | by PakAngels Inc

Link <https://credsverse.com/credentials/2e1bab37-b2dd-48ca-97e6-ed12c4ab6612>

07/2023 – 25/01/2024

IBM Back-End Development

Link <https://www.coursera.org/account/accomplishments/specialization/TDYLRGVH7PMV>

09/2023 – 15/10/2023

Secure Full Stack MEAN Developer | by EC-Council

Link <https://www.coursera.org/account/accomplishments/verify/Y4CYNCH2X2C9>

01/2023 – 30/01/2023

IBM Penetration Testing, Incident Response and Forensics

Link <https://www.coursera.org/account/accomplishments/verify/XASUQ76YXBEB>

06/2022 – 18/07/2022

Detecting and Mitigating Cyber Threats and Attacks | University of Colorado System

Link <https://www.coursera.org/account/accomplishments/verify/DELMF9ZG98Z3>

06/2022 – 25/07/2022

Disability Inclusion in Education: Building Systems of Support | University of Cape Town

Link <https://www.coursera.org/account/accomplishments/verify/GLNGBZET6ZPC>

05/2022 – 14/07/2022

International and Cross-Cultural Negotiation | ESSEC Business School

Link <https://www.coursera.org/account/accomplishments/verify/MY6BLZX28UFV>

15/07/2022

Kubernetes: Create Multi-App Cluster with Ingress & Logging

Link <https://www.coursera.org/account/accomplishments/verify/95N8ZSP3AGWX>

12/2021 – 05/05/2022

Google IT Support Professional Specialization

Link <https://www.coursera.org/account/accomplishments/professional-cert/6S3G22QPWJLH>

Link <https://www.coursera.org/account/accomplishments/verify/EXRNZLCBV2AR>

● COMPETITIVE HACKATHONS

07/11/2024 – 09/11/2024

ALLAM Hackathon | Saudi Data and AI Authority and SAFCP

- Collaborated with a team at SDAIA's hackathon in Riyadh to develop an AI-powered pipeline for real-time monitoring and analysis of Arabic social and electronic media.
- Delivered actionable insights for sectors such as defense, media regulation, and content filtering, demonstrating innovation and cross-disciplinary teamwork.

Other Competitions & Trainings

- Demonstrated interdisciplinary excellence by excelling in high-caliber competitions—including HRCA (robotics/STEAM innovation), MLSA CodeAir 2.0 (competitive coding), Kaggle Python (data modeling), and HITEC CodeWar Hackathon (time-based programming)—thereby honing advanced analytical, problem-solving, and teamwork skills.
- Augmented technical prowess through rigorous 5-day Cyber Security Awareness Training by NCSaEL Military College of Signals-NUST, mastering practical skills in malware analysis, network auditing, and real-time threat detection to build authentic, industry-ready cybersecurity expertise.

● HONORS & LEADERSHIP

02/2025 – 23/02/2025

Speech in Parliament | Youth General Assembly

- Selected from 200+ applicants to debate in the presence of the speaker of Punjab assembly of Pakistan on the topic, "Misinformation Endangers National Security".

10/2018 – 01/2019

Basketball Captaincy

- Spearheaded the formation and training of a competitive basketball team through strategic player selection, collaborative coaching sessions with school staff, and intensive skill development drills, demonstrating leadership in fostering team coordination, decisive decision-making under pressure, and analytical ability to optimize offensive/defensive strategies

14/12/2018 – 16/12/2018

MUN Delegate (DISEC)

- Spearheaded evidence-based policy proposals on disarmament through in-depth analysis of Indonesian geopolitical history, refining diplomatic communication & cross-cultural negotiation strategies to address global security challenges with analytical rigor.

● COMMUNICATION AND INTERPERSONAL SKILLS

Creative Problem Solving | Adaptability | Conflict Resolution | Emotional Intelligence | Active Listening

Leipzig , 01/03/2026



Humza Asrar Ahmed